

LinkShield AI: An AI-Powered Phishing URL Detection System

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1. Problem Statement :

The increasing number of phishing websites and online scams threatens users' privacy, financial security, and trust in digital platforms. Many users cannot identify fake websites, making them vulnerable to cyberattacks. LinkShield AI addresses this issue by using Artificial Intelligence to analyze suspicious URLs, detect phishing attempts, and promote safer online browsing, supporting SDG 16: Peace, Justice and Strong Institutions.

2. Objective :

- Develop an AI-powered system to detect phishing and suspicious website URLs.
- Analyze URLs and classify their risk level (High, Medium, or Low).
- Provide users with clear explanations and safety recommendations.
- Create a simple and user-friendly web interface using HTML, CSS, and JavaScript.
- Promote cybersecurity awareness and support SDG 16 by encouraging safer online practices.

3. Technology Stack :

HTML/CSS/JavaScript: To build a responsive and interactive web application.

n8n Cloud: To automate the AI workflow without building a backend from scratch.

Groq API: To provide fast AI-based phishing URL analysis.

Webhook: To enable communication between the website and the AI workflow.

4. Working :

- The user enters a website URL into the LinkShield AI web application.
- The frontend sends the URL to the n8n Webhook.
- The webhook triggers the AI workflow in n8n.
- The Groq AI model analyzes the URL for phishing indicators.
- The AI generates a risk assessment and security recommendation.
- The result is returned to the website and displayed to the user.

5. Implementation & Results :

Figure 1: Homepage of LinkShield

This is the homepage of LinkShield AI, where users can enter a website URL for analysis. The interface is designed to be simple, user-friendly, and responsive, allowing users to quickly check whether a website is safe or potentially malicious.

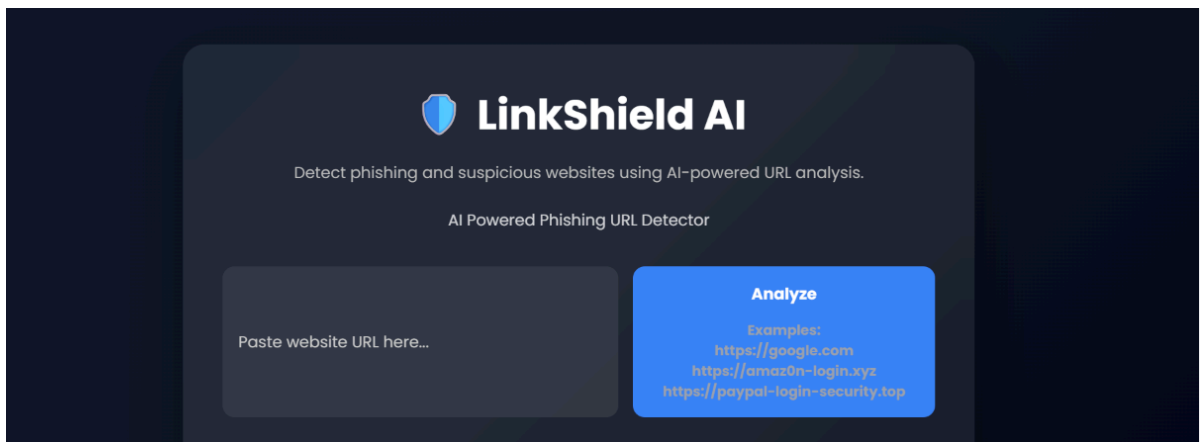


Figure 2: Phishing URL Detection

This screenshot demonstrates the AI successfully detecting a phishing website. After analyzing the entered URL, the system classifies it as High Risk, explains the reasons for the classification, and provides a recommendation to help users avoid potential phishing attacks.

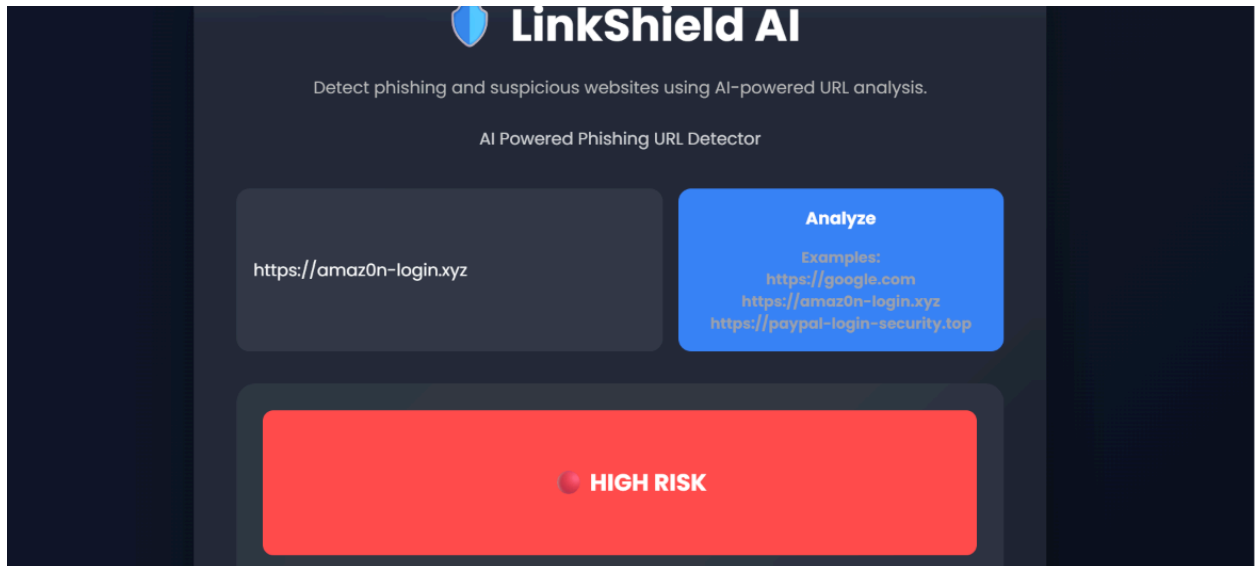


Figure 3 : AI Analysis & Recommendation

This screenshot shows the AI-generated analysis after scanning a URL. The system identifies the risk level, highlights suspicious characteristics such as brand impersonation or unusual domains, and provides a recommendation to help users avoid potential phishing attacks.

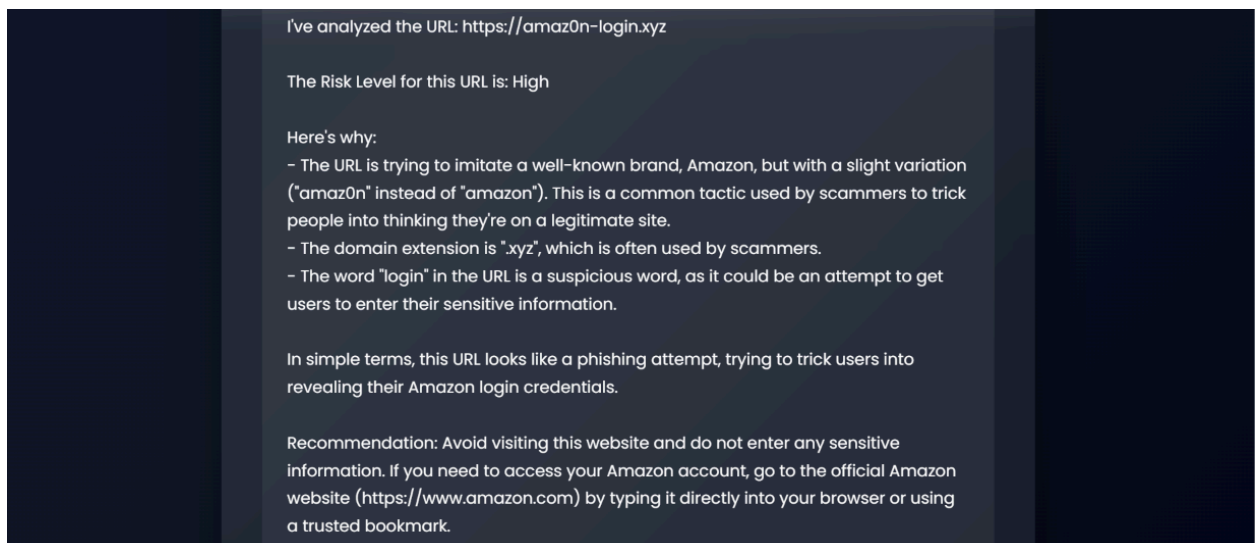
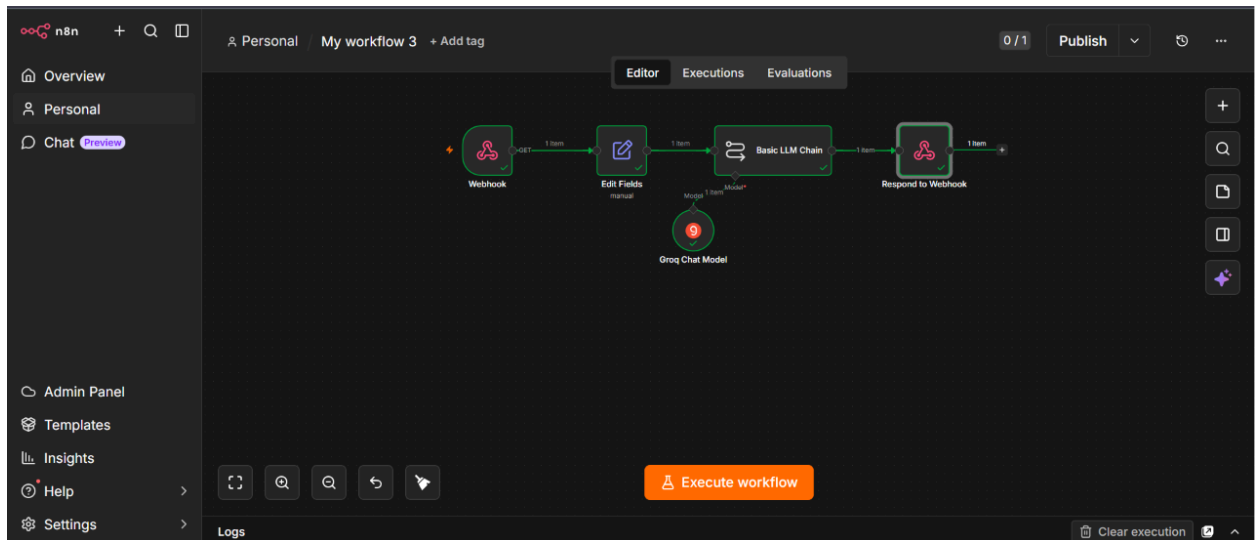


Figure 3: n8n Workflow Architecture

This is the backend workflow developed using n8n. It receives the URL through a webhook, sends it to the Groq AI model for analysis, and returns the generated risk assessment to the web application.



6. Future Scope:

- Integrate VirusTotal API and Google Safe Browsing API for more accurate phishing detection.
- Perform SSL certificate and domain age (WHOIS) verification for enhanced security analysis.
- Develop a browser extension to warn users before they visit suspicious websites.
- Maintain a scan history and generate security reports for users.
- Expand the system into a mobile application with real-time URL scanning and alerts.

7. Conclusion:

LinkShield AI successfully demonstrates how Artificial Intelligence can be used to identify phishing websites and improve online safety. By analyzing suspicious URLs and providing clear risk assessments with recommendations, the system helps users make safer browsing decisions. This project contributes to SDG 16 (Peace, Justice and Strong Institutions) by promoting cybersecurity awareness and building trust in digital platforms.

8. Project link: [linkshieldai](#)